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Clients and carers

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Clients and carers: Healthcare professionals' roles in medical device development processes in SMEs

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ABSTRACT

Purpose of the article: With the call for patient-centred healthcare, designers and healthcare professionals collaborate increasingly, as both health and design researchers advocate closer engagement between the two. Yet the existing literature is scarce in describing the actual roles taken up by healthcare professionals in these collaborations. Addressing this gap, this article investigates the participation of healthcare professionals in medical device development in SMEs.

Materials and methods: The research context is Ankara, Turkey, where most design and development activities in medical sector are geared towards export substitution, undertaken with limited resources. The analysis is based on data gathered through semi-structured interviews with 32 professionals, including design professionals, managers, physiotherapists and engineers involved in the design of around 25 medical devices.

Results and conclusion: This article identifies four main roles that healthcare professionals take in medical device development: business partners, expert users, mediators, and medical professionals. These roles entail diverse responsibilities with various objects of collaboration with designers, making significant contributions to medical design.

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Introduction

Scholars in the field of health have claimed that interdisciplinary collaboration between healthcare professionals (HCPs), and design teams should be enhanced to address healthcare problems that require innovative solutions. They argued for creating opportunities where HCPs' expertise can be incorporated into the design of medical devices and technologies (see e.g. Zhou, Li, and Li 2021; Glasgow et al. 2018). Some studies suggested that skills and knowledge for collaborative work should be integrated into the professional training of HCPs, e.g. via utilising design thinking (van de Grift and Kroeze 2016) or through joint innovative education programmes on healthcare technologies (Glasgow et al. 2018; Liu 2021). Others have questioned the limitations of the conventional linear product development process widespread in the medical industry, where HCPs' expected role is limited to outlining the functional and cost requirements of products. Instead, scholars proposed open innovation and business models, which encourage participation of all relevant stakeholders, also including scientists, operators and patients (see e.g. Clemensen et al. 2007; Davey et al. 2011; Formicola et al. 2021).

The field of design for healthcare, in turn, has built on cross-disciplinary experiences in design of medical devices, where educators have similarly argued for the need to equip designers and HCPs with each other's core competences (e.g. Goossens, Lange, and Kleinrensink 2004; for an overview, see Noël and Frascara 2016). Design for health research today places considerable emphasis on incorporating user experiences into design through human-centred approaches (e.g. Coleman and West 2011; West et al. 2014; Page 2018; McCarthy, Rodríguez Ramírez, and Robinson 2017), sometimes in explicit contrast with engineering-led approaches to medical device development (Bush 2015; Hargraves 2018). The focus moved towards patient-centredness, and within that, improving patients' autonomy in managing their own health (see Smits et al. 2022).

Accompanying the emphasis on patient involvement, the growing literature has clearly recognized the benefits of interdisciplinary collaboration amongst HCPs, designers, and other stakeholders in the design and development of medical devices. A comprehensive review in 2015 showed that designers and HCPs are engaged in equal numbers in published research on design for health (Chamberlain et al. 2015). However, the existing literature is scarce in in-depth accounts and systematic descriptions of the specific roles which HCPs play in such collaborations. We especially need to know more about practices carried out in organizational contexts outside state-of-the-art research environments, whose good practices dominate existing literature. To address this gap, in this article we present a framework for identifying HCPs' roles in design and product development processes. Our findings

are based on interviews conducted with designers and other professionals involved in medical device development in one such context: medical device design and product development activities in small and medium enterprises (SMEs) in Turkey. The article starts with a review of both health and design literatures on designer–HCP collaborations, followed by a description of our research context and research design. We present our findings in the form of a classification of roles, responsibilities and objects of collaboration.

HCPs in design-led medical device development

Owing to the increased appreciation of user-centred design approaches in medical device industry, the role of HCPs as expert or professional users of medical devices has gained prominence in the last two decades. Studies have proved that it is required to identify the relevant clinical applications and users correctly early in the process, so that accurate information about user requirements, use context and scenario, and existing problems can be obtained (Martin et al. 2012). Such information is valuable to designers of medical devices that offer ease of use, improved safety and effectiveness, reduced risk, and simpler maintenance (Bitkina, Kim and Park 2020). Smith et al. (2019) argued further that including HCPs, specifically clinicians, in the early stages of medical device development as key stakeholders enables them to take further roles, and contribute to 'needs-driven problem solving': first, clinicians have the experience in patient care, the 'state-of-the-art' clinical and diagnostic knowledge, and access to various resources (such as the literature and professional societies) to define the scope of a clinical problem. Second, clinicians can bring insights regarding user needs, from HCPs to end users such as patients and their families. A professional user's connections to a community of peer experts can provide even wider access to use-related information (Afuah and Tucci 2012). Third, clinicians can advise manufacturers on, for example, the value proposition of the medical device, marketing strategy, pricing strategy etc.

Paralleling the third point, scholars have increasingly argued that HCPs can undertake more active business-related roles in medical device innovation and entrepreneurship (Lettl, Herstatt, and Gemuenden 2006; Katila et al. 2017; Janardhan and Janardhan 2022). Such recent studies confirm that many medical device companies refer to HCPs, specifically medical doctors, on issues such as addressable gaps in the market, and device preferences. Accordingly, companies are advised to establish permanent collaborative engagements with medical doctors by inviting them to the company's scientific advisory board, where they can provide sustained guidance to design teams – including keeping the teams focused on the impact of new products on patients' lives rather than technological novelty.

The literature on design for health offers similar arguments through case studies that involve HCPs in design-led development. In user-centred activities, the input from HCPs, including doctors, nurses and therapists, are used to complement the participation of end-users and their carers to help contribute to the designers' understanding of patient needs (Daemen et al. 2015; Aydin et al. 2020) as well as other experiential and institutional factors (Langley et al. 2018). In the case of personalized assistive devices, this refers to clinicians working with individual patients to identify constraints and opportunities (De Couvreur and Goossens 2011; Henley and Rodríguez Ramirez 2018). In other cases, design process is approached more holistically from a co-design perspective, where iterative testing and co-design activities involve diverse stakeholders, from technical professionals to governmental and civil-society actors, also including patients, HCPs and designers. Moody (2015) lists HCPs' contributions to such projects as providing functional and risk assessment on early prototypes, and participation in decision making regarding the prioritization of diverse concerns raised during process. Papers using co-design approaches indicate the need to align differing disciplinary interests between actors, including HCPs' focus on correct treatment and good care (Coleman and West 2011; Moody 2015; Ankeny 2020).

Our review points to two conclusions and related research gaps. First, the value of HCPs' involvement in medical device design has been acknowledged widely. Nevertheless, the evidence of long-term and sustainable involvement in practice is scant, and apart from the professional user role, the extant research mostly discusses the roles that doctors *can*, rather than *do*, play. Second, the design literature, in a similar vein, provides arguments for extensive involvement of HCPs in medical device development, and design for health case studies report engagements with HCPs, not least in the form of co-design activities that involve diverse stakeholders. However, the existing work does not provide a comprehensive description of what HCPs contribute to design-led collaborations, and there is lack of interest in everyday medical device development in the industry as opposed to state-of-the-art experimentation in connection with research institutions. Our research design aims to address these gaps by examining the roles and responsibilities HCPs *do* undertake in medical design projects implemented by SMEs.

Research design and context

In this article, we investigate the *dimensions of collaboration between HCPs and designers in medical device development processes*. To this end, we designed an interpretive qualitative study to recreate the design process, its steps and relevant design considerations from the perspectives of diverse professionals

involved in medical device development in SMEs. We adopted semi-structure interviews to gain an in-depth understanding of professionals' experiences (Gubrium and Holstein 2009). In total, we conducted semi-structured interviews with 32 professionals.

Prioritizing the design perspective, the recruitment of participants was initiated by reaching out to design graduates working in the medical industry. A list of potential participants was formed through a review of websites and published reports about two major medical industry clusters and university research parks in Ankara. The sample includes 18 design professionals (industrial designers, design managers, and usability researchers), whose diverse expertise across several design projects with manufacturing companies and HCPs were useful for providing both wide and in-depth insights regarding such collaborations. We used snowballing as our second sampling strategy to locate and interview other professionals (Parker, Scott, and Geddes 2019), where design professionals were asked at the end of interviews to provide contacts in the medical device industry with whom they formerly collaborated. This established a secondary group of participants, including 6 company executives, 3 physiotherapists, and 5 R&D engineers (see Supplementary material A). Recruiting HCPs for interviews proved particularly difficult (Moody 2015; Langley et al. 2018). Company executives were interviewed due to the central role they played in product development activities in their respective SMEs.

Our participants were based in 16 businesses in Ankara, Turkey, including manufacturing SMEs, technology start-ups and design consultancies. In Turkey, medical device sector is considered a growing sector with thousands of manufacturing companies. Around one-third is based in Ankara, where there is an established medical device industry with a large number of SMEs, each specialized in specific medical equipment, and which rely on a small group of design consultancies for design services. Research and industry reports indicate the limitations of the sector regarding high-technology products, and the need for better integration of design and marketing activities (AKA 2013; Köse and Çifter 2018). This parallels findings from research on medical device SMEs in other countries, where the SME's limited resources inhibit them from employing user-centred approaches extensively and effectively (Martin et al. 2012; Smith et al. 2019).

The interviews concerned the design and development of around 25 medical products, including surgical devices such as aspirators and biopsy guns, therapy equipment such as game-based rehabilitation systems and phototherapy units, hospital furniture and electronic systems, and various laboratory devices. A majority of these devices were developed for export substitution for Turkish and surrounding developing markets. We designed a

topic guide (Arthur and Nazroo 2003) that allows the participants to walk the interviewer through as many projects as they find relevant. The guide consisted of three sections: The first section collected information about the participant and the company. The second section elicited a narration of steps, collaborations, and considerations for each project where there was a designer-HCP collaboration, starting from conception to implementation. The third section asked for reflections on the opportunities and barriers in collaborations (see Supplementary material B). In most interviews with designers, they shared visual materials, e.g. sketches, CAD models, and prototypes.

Interviews lasted 45 to 60 min, except interviews with more experienced designers, which lasted 90 to 120 min due to the large number of projects reviewed. All interviews were audio-recorded with the informed consent of participants and fully transcribed. Transcriptions were analyzed using NVivo analysis software. Data was coded in two cycles. The first cycle had an in-vivo, simultaneous coding approach, while the second cycle adopted focused coding that concentrates on (i) the design process, steps, and activities, and (ii) various professionals' roles, responsibilities, considerations, and values (Saldaña 2009).

Interviews were conducted in Turkish, and quotes used in this article were anonymized and translated.

HCPs as business partners

As business partners in design projects, HCPs undertake responsibilities as clients and advocates. In this role, their contributions are mainly to the business, e.g. identification of business opportunities and potential stakeholders, aiding in promotion, and indirectly to design process, most importantly by shaping product specifications, as explained in the next two sections.

As clients

A significant number of projects we explored were initiated by HCPs such as medical doctors and technicians who acted on a perceived business opportunity. Some of these projects were rooted in the professionals' own needs at the clinic. One example was an innovative patient chair that supports specific postures that were not offered by existing products (Designer A). In addition to making the operational procedure less arduous for both HCPs and patients, the aim was also to make it more efficient, and thus improve the doctor's patient throughput. In such cases based in the clinic, it was always the doctor as the client who provided the brief in the form of technical specifications. In the example of the patient chair, the specifications described the posture and placement of the nurses and patients on and around the chair; in another case of a surgical cutter project (Designer L),

this concerned the characteristics and distribution of surgical cuts the cutter was expected to deliver.

Others were export substitution projects where a medical professional identified a market for producing cheaper, domestic alternatives to medical products that were currently provided solely by global brands, such as screening kits, therapy equipment, laboratory devices, and their disposables. Such projects often started with reverse engineering the original, branded product, then developing an alternative with limited technical and aesthetic differentiation.

As typical in SMEs (Martin et al. 2012; Smith et al. 2019), both types of projects are often undertaken with both limited funds and limited product development and manufacturing expertise. Some take advantage of public funds for research and development. According to design consultants who had experience with such projects, many start with a highly technological focus and end up with what Designer A called 'a gadget' – a prototype that lacks an understanding of use, and possibly of mass manufacture and of the market.

The client then hires a designer for mostly aesthetic and simple usability improvements. Designer C related that most such clients consulted them in response to requests from distributors in other countries for better-looking designs. For example, for Physiotherapist C, who is also an entrepreneur, negative client responses about the user interface and the overall size of the product were what triggered them to think about hiring an industrial designer; but ultimately wanted a product that is 'more beautiful, aesthetic and functional,' similar to 'equivalent products in Europe.'

In these cases, designers often end up contributing more extensively, from material selection and ergonomic considerations to introducing user needs, including patient advocacy in relation to issues such as trust, psychological safety, and social stigma, on to advising on the business model. This indicates a potential for earlier and more holistic collaborations between design teams and HCPs. Indeed, some of our interviewees were particularly appreciative of medical doctors' contributions to such collaborations. Executive C, for instance, argued that 'some doctors are both really good doctors and have a head for [product development],' where they can guide a design team with their command of medicine, treatment, technology as well as business.

As advocates

HCPs play critical business-related roles also when they are not business owners. According to Executive C, it is common advice for start-ups to get 'champions' from the related technical field, in this case, medical professors

that are recognized as authorities in their field, whose backing can significantly strengthen a project's credibility. In this article, we prefer the term 'advocate,' since their support might be provided out of altruism or genuine interest, as well as paid for. Our participants described the various ways a prominent doctor's advocacy for a project can be demonstrated to the relevant medical and/or industrial community: The doctor can use the medical device in their own research and publish the results; they can be hired as paid speakers in academic and industry conferences; and more informally, their names can be mentioned in business-related conversations or in promotional documents and correspondence. Customers such as hospitals, manufacturers and distributors also have their own consultant doctors, and a well-known professor can be helpful in reaching and convincing those. Lastly, an advocate can help a team navigate the field by introducing them to relevant individuals such as specialized medical professionals or industry leaders, who might otherwise be difficult to locate for start-ups as well as design teams looking for information and feedback.

One story, narrated by Physiotherapist B, described how another colleague put her in contact with a designer-led start-up that needed a physiotherapist's support, motivated by the desire to help foster medical technology development in Turkey. Their collaboration started with the physiotherapist aiding the design team with her knowledge of physiology as well as experience with user needs. Later, her help was enlisted when an expert's signature was required for a patent application. Indeed, doctors as advocates can take on tasks in relation to formal processes in technology development. Another of these is overseeing and undersigning clinical certification processes for medical devices. A third one is related to the writing of technical specifications for medical device tenders for hospitals. It is crucial for manufacturers to have the knowledge, or better yet, a shaping influence over what is written in a technical specification document, which is often shaped by 'opinion leaders', i.e. well-known doctors in a specific field whose advice others would seek whenever new equipment is being purchased. Having contact with an opinion leader is therefore of critical value, and can have a more direct bearing upon product development since both clinical certification and official technical specifications shape both the process and the end product.

HCPs as expert users

Being an expert user was identified above as the most established role of HCPs in the literature. The designers we interviewed also considered it part of the 'textbook' design process to involve HCPs as expert users. Accordingly, observations in medical procedures, interviews, and early user tests with HCPs were often considered requirements for good design.

In addition to issues of medical efficacy, HCPs comment on a wide range of use-related issues. Comfort, especially in relation to work posture, was central to many projects. The design problem here is formulated as a user experience problem rather than mere ergonomics, so that designers address different dimensions of experience – such as, in the patient chair project, the discomfort of women nurses during the procedure where the seat design brought their breasts too close to men patients' faces. In other projects, it concerned the patient's psychological safety and trust in the proposed technology. In yet others, there were considerations about various use scenarios, tacit rules in use contexts, as well as habits and competences of different actors involved.

Many participants explicitly differentiated the role of surgeons, nurses, physiotherapists and medical technicians as expert users, from the role of general practitioners and consulting doctors as informants who possess technical knowledge of the treatment but do not deliver it themselves. According to Designer E,

> most devices, like a ventilator, have nothing to do with the doctor. The nurse must comment on those. How the technology works is for the doctor to [know]. That's what the doctor needs to discuss with the engineer, whereas the designer typically gets his information from the nurse.

For him, it is the doctor that has medical knowledge of diseases and related interventions, whereas the designer is interested in knowhow. This view can be compared to other designers' arguments cited above that engineering-led collaborations produce 'gadgets.' Multiple designers and physiotherapists we interviewed contrasted doctors' medical knowledge with physiotherapists' hands-on understanding of user needs, characteristics and experiences. Designer P called it the 'human touch;' Physiotherapist A emphasized their prolonged and intimate interaction with patients, which allows them to identify weaknesses and offer specific treatments; and Physiotherapist B mentioned perceptiveness and patience as defining features of her profession, where they need to care for individual patients.

Similarly, nurses were considered indispensable as expert users for most medical devices in clinical environments. In one project, nurses gave regular feedback to the design team about the usability issues they encountered during the test period of a product, also mediating the needs and opinions of doctors and patients involved. In another case, the manufacturer had a permanently employed ex-nurse in the marketing team, so that the design teams could refer to her previous experiences whenever they needed input.

HCPs as mediators

In addition to being business partners and expert users, a third role that HCPs undertake in design projects is mediating between, first, medical and

technical domains necessary for medical device development, and second, designers and other actors involved in the use of the products, including other HCPs, patients and caregivers.

Between medical and technological domains

The designers we interviewed consulted books, articles, surgery videos, internet sources and technical documentation for existing devices to gain a basic understanding of the medical issues they worked on, be they anatomic geometries, surgical procedures or treatment protocols. This helped simplify the medical-technical dimension of the design problems: Biopsy gun design is about loading a spring; pap-smear is a question of cervical geometry; an LED phototherapy device is essentially a lighted transparent box. Designers also talked to HCPs for this purpose. Designer Q described his experiences during his graduation project on chest physiotherapy, where he visited a professor to learn about how the treatment needs to be applied:

> I learnt that how hard you clap matters only when I visited [the professor]. I hadn't been able to find any academic sources on that. I asked her how she does it. She said, it's years of experience, we got used to it. She showed it, 'this fast,' but I asked whether there's anything [more concrete]. Then she found a physiotherapy source from the bookshelf, and said: 'Oh, yes, it says here, look, you need to clap with 3.5-5 hertz.'

It is important to note that the designers we interviewed found it especially valuable when HCPs were not just relaying information, but co-operated in a way that helped them achieve a better overall understanding of the physiological and clinical issues, as in the quote above. This can include locating relevant sources, interpreting formal descriptions with reference to tacit knowledge of clinical practice, and translating terminology.

Terminology is especially salient. Designer Q above told us how he used to have difficulties understanding the medical language used by HCPs, and record conversations only to look up terms later. Designer A added that having acquired a knowledge of relevant medical terminology in earlier projects, now helps her communicate more convincingly with doctors. Physiotherapist A, who consults for a medical robotics company, described similarly that one of her major tasks has been to act as an interpreter between the design team and clinicians who participate in trials or are under training as users: Simple translation issues, such as whether certain muscle movements should be represented with a negative number (engineers) as opposed to a lower positive number to indicate low patient mobility (physiotherapists) can inhabit mutual understanding and lead to delays.

This further means that mediating HCPs not only translate medical knowledge and terminology to designers, but also convey how the design team

thinks and works to stakeholders from the health sector. One example by Physiotherapist C points towards the need for health professionals who collaborate with technology developers to develop a sense of what is technologically possible – in his example, to what extent a complex shoulder movement can be feasibly reproduced by motors. Lastly, advocating doctors may be considered to take on a similar role when they translate technical features into clinical benefits through their scientific work and presentations to health professionals who are potential customers or users (Executive B).

Between designers and others

In our interviews with designers, we often ended up talking about issues of access to fieldwork. As noted above, clients with limited sources may insist to go ahead with their own knowledge of clinician and patient needs. This might be communicated to designers in the form of reports, which can consist of detailed marketing reports in wealthier companies; otherwise, it is delivered as informal communication of functional expectations. Companies might also choose to test early prototypes with HCPs without the participation of design teams, then ask directly for revisions.

In other cases, medical doctors as clients or expert users might choose to talk in the name of nurses' and technicians' needs. Most often, though, it is the patient's perspective that is mediated by HCPs. Patients were excluded from the majority of the design processes we observed, particularly if they are secondary users – e.g. at the receiving end of the diagnosis or treatment by the health professional using the medical product. This has been underlined as a problem in literature (Alsos and Svanæs 2011; Angell, Keitsch, and Sigurjónsson 2016). In our fieldwork, we noticed that this was also an issue where the patient is the primary user, and HCPs secondary, as in home treatment devices and prostheses. In relation to a tracking device for elderly patients, Executive C mentioned their reluctance to have contact with patients themselves, citing their business-to-business approach with hospitals, and relied instead on nurses, who would be tasked to pass on not only their own preferences as users, but also any feedback from patients and their caregivers.

In addition to conveying feedback, HCPs can also have characterizations of patients that colour their feedback. Designer P's impression, for example, was that HCPs tend to be more pessimistic about what patients are capable of. The HCPs they collaborated with for an inhalation device design believed that the patients – typically elderly patients from low socio-economic status and with low literacy levels – had neither the digital literacy nor the health literacy required to effectively employ high-end, connected devices. Such observations based on personal experiences at the clinic can shape design

approaches and business strategies. Here, the HCP's patient characterization led both design and marketing efforts to focus more on patients' adult children as caregivers, who will purchase and use the device together with their parents. A doctor's distrust in patients can also lead to solutions that limit device affordances for users. Designer F's design solution for an external fixator design for healing broken bones, for instance, required a key to prevent patients from manipulating the device.

HCPs' role in mediating others' perspectives becomes more pronounced when they are also gatekeepers of the clinic. It is typically HCPs who allow designers into intensive care units and operation theatres for observations, and request their assistants and nurses to help designers with interviews and user tests. They are also indispensable for recruiting patients, who can be simply difficult to locate, along with the strict regulations that govern patient contact. It also necessitates clinical expertise and experience to discern patients. In Designer O's experience, good collaborations require patients that have the correct diagnosis, as well as the physical capacity and the temperament, which can only be assessed by HCPs.

HCPs as medical professionals

Last by not least are HCPs *qua* medical professionals, where their participation in new product development is motivated and led by their interest in clinical good. Distinct from their roles as expert users, HCPs in the role of medical professionals align the design process to their own professional interests and objectives, namely research and clinical care.

As carers

HCPs' roles as carers can become prominent in a project. According to Designer O, HCPs also manage the extended relationship of the design team with the patient. Patients are not only recruited with attention to their diagnosis, but also with an eye to their specific care requirements. These may be about how their illness will respond to the treatments provided by the device under trial, both preventing harm and improving quality of life. These may otherwise be about personality traits – whether a patient is nervous or unable to relax – or even whether they would be able to find time for long-term trials. All such considerations can only be observed from the view of a long-term carer – such as a nurse or a physiotherapist as opposed to a general practitioner, as in our participants' comparisons above. In Designer O's experience, the physiotherapist makes such observations, and even intervenes, for instance by assigning regular exercise to the patient before they are ready for trials.

In our fieldwork, we observed a close relationship between a HCP's understanding of specific clinical benefits and new product development activities. Executive B was adamant that medical device development should build on specific clinical goods, based on the hands-on experience of care. Physiotherapist C similarly told us that a patient's difficulties with specific everyday tasks, such as standing up from a chair, can require improvised solutions to help them, which can then trigger new product ideas. Physiotherapist A provides a concrete account from her experience with walking therapy robots:

> Is this product useful from the health professional's perspective? Is it meaningful? Does it provide a benefit? … They need someone to ask these things. … For example, they ask me: Should we make the patient walk backwards? I mean, would that be helpful? … My answer is, then, that walking backwards can help the patient improve their balance, by making the task harder for a patient that is already improving. But we cannot use it for a new, 'acute' patient. … Or I suggest them [specific games] in order to improve the patient's hand coordination. … I suggest them that the game should have a purpose, the patient should not get bored, we need to have objectives in place: Do we want the patient to take larger steps? Let's design a game for that, like jumping from one stone to the other.

As seen in the quote, HCPs help translate design ideas to clinical goods, and conversely provide concrete cases of clinical benefit on which the design team can capitalize. Such ideas need to be checked against existing literature, such as whether extant work has explored the health benefits of walking backwards. If not, this can also result in new research.

As researchers

In several interviews, we have heard about HCPs in their roles as researchers. We have already mentioned above that companies collaborate with doctors who promote their products to medical communities through academic presentations and publications. Here, the relationship is mutual insofar as a medical researcher's collaboration with a company and its design teams in turn contributes to improving the former's scientific output. We already noted that HCPs are indispensable for recruiting patients for user trials during and after product development processes. Furthermore, for medical device certification, companies collaborate with HCPs to design and run clinical investigations, with responsibilities such as patient sampling and selection and application of standardized tests. This also includes doing literature reviews about existing clinical applications. Physiotherapist A argued that, even though clinical assessment methods would be known to anyone working in a clinic, a command over literature sets those that work in an academic setting apart in terms of how they can contribute: 'For academics, measurement and evaluation are at the forefront, so: Can we measure it

Table 1. HCPs in medical device design.

Roles	Responsibilities	Object of collaboration
Business partners	HCP is **client** for designer, recognizes business opportunities, prescribes product specifications, steers business plan.	Collaboration is around a **business problem** in relation to new business opportunities, a market (buyers), competitors, as well as regulations and certification processes
	HCP is an **advocate** for the design team, strengthens their credibility, provides access to potential stakeholders, oversees and undersigns formal processes.	
Expert users	HCP is an **expert user**, providing experiences and insights from the clinic when consulted in fieldwork.	Collaboration is around a **user experience** problem, where HCP represents the user.
Mediators	HCP is a **mediator between medical and technological domains**, sometimes between the design team and other HCPs.	Collaboration is around a **technical problem** that can be addressed by access to correct terminology and/or good knowledge sources as well as developing a shared understanding.
	HCP is a **mediator between designers and other stakeholders**, speaking in the latter's name in their absence.	Collaboration is around a **user experience** problem, where HCP represents other user groups.
Medical professionals	HCP is a **carer** in the clinic, relying on clinical knowhow, responsible for patients' wellbeing.	Collaboration is led by **clinical goods**, with an eye to health improvements to specific patients.
	HCP is a **researcher**, focused on design, conduct and dissemination of research activities, including literature reviews and clinical tests.	Collaboration is led by a **research problem**, i.e. contributions to clinical science, often formalized in the form of usability trials as part of certification processes.

reliably, in a repeatable fashion? Can we emphasize the scientific aspect and present it to academics?'

Concluding discussion

Our findings outlined several distinct but intermeshed ways in which HCPs contribute to medical device development and design within SMEs in the context of medical industry in a developing country, where the majority of related activities are geared towards export substitution. Table 1 summarizes our findings in the form of a framework of roles, responsibilities and objects of collaboration.

Amongst the roles and responsibilities outlined in this article, designers we interviewed consider HCPs' two roles to be particularly critical in design projects. The first is their role as mediators, where design teams rely on HCPs' both knowledge and human resources to enhance their understanding of the medical issues they focus on. It was not the medical authority the designers sought in their engagements with HCPs as mediators, but that the latter can suggest books and articles, provide surgical videos, and grant access to hospital spaces to observe, patients and other HCPs to talk to, so

that the designers can build their own understanding of the issue at hand. The second was HCPs' role as carers, where they could contribute to the process whereby a design idea's clinical worth is assessed, utilizing tacit knowledge developed in the clinic. This is parallel with observations from co-design processes where HCPs' interest in clinical good was considered a defining feature of interdisciplinary collaboration in health (Coleman and West 2011). This is why as carers – and as expert users – designers held clinicians in higher regard, as opposed to general practitioners, who lacked the knowhow.

Recently, it has been suggested that the successful development of a medical device needs not only the external contributions of HCPs as users, but also close interdisciplinary collaboration between HCPs and designers as a team (Yoda 2016). In this research, none of the processes we have observed were based on thorough interdisciplinary collaboration as advised in literature. Integration of disciplines and focus on a shared problem lie at the core of interdisciplinarity (Repko 2012). In the projects covered in the interviews, the focus on a shared problem was evident, while integration was missing. The reasons would vary among projects, e.g. due to limited resources or expertise of the SMEs or designers' lack of experience with or interest in collaborative methods. It is however clear that HCPs contributed ideas and so shaped design decisions in their various roles, for instance, defining product specifications as clients, defining the clinical good as carers, and interpreting and reflecting external stakeholders' perspectives as mediators. Moreover, regardless of the level of integration, professional development of each team member is one of the key success criteria in collaboration (Levi 2017). This criterion seems to be fulfilled in the projects we investigated, where designers improved their understanding of medicine, and HCPs could engage in better care and more research through the collaboration.

Besides summarizing the outcomes of this research, Table 1 also reveals directions for future research. First, we argue that the table should be expanded through fieldwork in other geographical contexts. Second, in this study, we started tracing the existing projects through the designers. Future research that conducts further qualitative studies with HCPs themselves would enrich the framework provided here.

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